

2024

Time : $1\frac{1}{2}$ hours

Full Marks : 40

*Candidates are required to give their answers
in their own words as far as practicable.*

The figures in the margin indicate full marks.

Answer both sections as directed.

Section-A

1. Answer any **four** questions of the following :

$$6 \times 4 = 24$$

(a) Prove that the identity element in a group is unique.

(b) Let G be a group and H a non-empty subset of G . Then H is a subgroup of G iff $a \in H, b \in H \Rightarrow a.b^{-1} \in H$.

- (c) To find the necessary and sufficient condition for a non-empty subset W of a vector space $V(F)$ to be a subspace of $V(F)$ is a $a, b \in F$ and $\alpha, \beta \in W \Rightarrow a\alpha + b\beta \in W$
- (d) The intersection of any two subspaces of a vector space is a subspace.
- (e) If A, B be two non-singular matrices of the same order, then $(AB)^{-1} = B^{-1}A^{-1}$.
- (f) Find the inverse of the matrix

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$$

Section-B

2. Answer any four questions of the following :

$$4 \times 4 = 16$$

- (a) Prove that $\alpha = \{1, -1, i, -i\}$ is a group with respect to multiplication.

- (b) Prove that every field is an integral domain but converse is not necessarily true.

- (c) Let $V(F)$ be a vector space and O be the zero vector of V . Then

(i) $a \cdot O = O \forall a \in F$

(ii) $a \cdot \alpha = a \cdot \beta \Rightarrow \alpha = \beta \forall a \in F$

$$\alpha, \beta \in V, a \neq 0$$

- (d) Define vector subspace. Show that W of ordered triads $(a_1, a_2, 0)$ where $a_1, a_2 \in F$ is a subspace of $V_3(F)$.

- (e) Solve the following system of equation by matrix methods :

$$x - 2y + 3z = 5$$

$$4x - 3y + 4z = 7$$

$$x + y + z = -4$$

(f) Find the eigen value of the matrix

$$A = \begin{bmatrix} 9 & -1 & 9 \\ 3 & -1 & 3 \\ -7 & 1 & -7 \end{bmatrix}$$

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